Detailed description of the design including the CONTROL treatment and the POLICY treatment. Please particularly outline the intervention in the POLICY treatment in detail.

[CONTROL]

Burning fossil fuels leads to considerable economic damages, such as due to climate change. The economic damages from climate change are commonly summarized by the social cost of carbon (SCC). The SCC defines how much society should pay to avert future damages caused by climate change, such as increased floods and droughts that are expected to intensify over the coming decades, by adding up all the quantifiable costs and benefits of emitting one additional ton of CO2 far into the future.

Due to the range of economic damages from climate change that will unfold over centuries into the future, the former chief economist on the White House Council of Economic Advisors, Prof. Michael Greenstone, has termed the SCC as the “most important figure you've never heard of”. The US

Environmental Protection Agency (EPA) has recently published an updated estimate of the SCC. According to the EPA’s most recent guidance, each extra emitted ton of CO2 causes economic damages of around USD 190.00 per ton of CO2 on average.

A key policy proposal to address climate change is to put a price on the use of fossil fuels, either by means of a tax or by introducing emission markets, such as the cap-and-trade program in California or the European Union Emissions Trading Scheme. The revenue raised by this carbon price could be directly returned to citizens, such that most extra costs are alleviated while retaining the incentive to buy goods and services that are less polluting.

Recently, scientists have presented new evidence on the effectiveness of emission markets to reduce carbon dioxide emissions. The cap-and-trade program in California has sizably reduced carbon dioxide emissions. Reductions in emissions driven by the European Union Emissions Trading Scheme, with its higher carbon prices, have been even larger. Carbon pricing is thus effective in reducing climate damages that will predominantly hit future generations.

[POLICY INTERVENTION]

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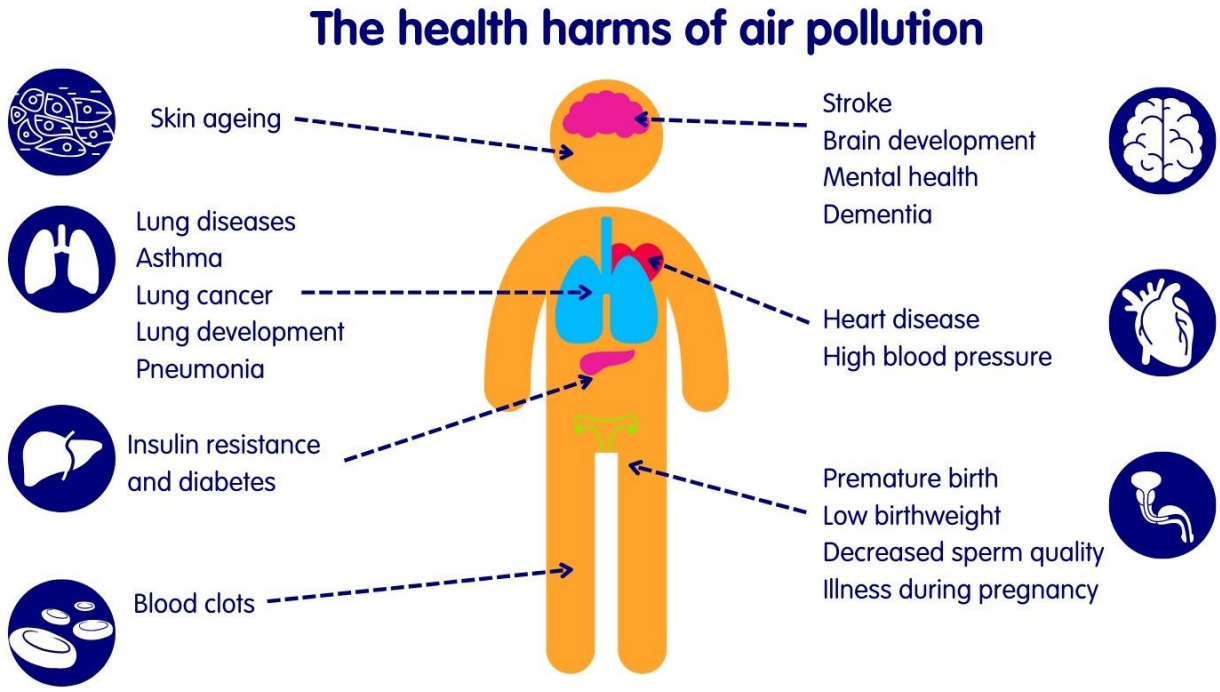
However, burning fossil fuels not only leads to climate change that affects humans in centuries to come, but also leads to air pollution. Scientists have linked toxic air pollutants, caused largely by the burning of fossil fuels, to respiratory problems like asthma and bronchitis, cardiovascular diseases, lung cancer, premature death, and reduced economic productivity (see pictures below). Researchers from the University of Chicago, led by Michael Greenstone, have estimated that air pollution is cutting life expectancy globally by, on average, almost 2 years today. According to Prof. Michael Greenstone, air pollution is therefore today "the greatest external threat to human health on the planet".

Air pollution is

"the greatest external threat to human health on the planet".

(Image credit: US Environmental Protection Agency (EPA))



(Image credit: Global Action Plan).

Recently, scientists have presented new evidence that the cap-and-trade program in California has—as a co-benefit—incidentally reduced the emission of toxic air pollutants. Similar findings have been made for cap-and-trade programs and fuel taxes in Canada and Europe. Thus, carbon pricing not only reduces long-

term climate damages but also delivers immediate local public health benefits and productivity gains by reducing toxic air pollutants.

Pricing fossil fuels using carbon prices delivers immediate public health benefits by reducing air pollution.

Detailed description of the outcome variable "real-world support for a price of carbon" and especially its real-world impact.

The US Environmental Protection Agency (EPA) has recently published an updated estimate of the social cost of carbon (SCC), according to which an extra emitted ton of CO2 causes economic damages of around USD 190.00 per ton of CO2 on average. If introduced as a price on the emission of CO2, such as through a cap-and-trade program or a tax, an SCC of USD 190.00 would add, for example, an added cost of about USD 1.60 per gallon of gasoline. This estimate of social costs does not include other harmful effects of the use of fossil fuels, such as health and productivity damages from toxic air pollutants.

There is disagreement in society on how high an appropriate price on the emission of CO2 should be. A carbon price that the US government should introduce may differ from the SCC for a number of reasons. These include concerns about international competitiveness, or additional co-benefits of carbon pricing due to the reduction of toxic air co-pollutants.

We would now like to get your view on how high a federal price on the emission of CO2 should be in the United States. The results of this survey will be communicated to a Chief Environmental Economist working in the White House. We therefore kindly ask you to carefully consider your response. Please state your view

on the appropriate added cost per gallon of gasoline, with options ranging from USD 0.00 to USD 5.00 per gallon, by moving the slider to the position you deem appropriate.

[SLIDER]

[USD 0.00 added cost per gallon of gasoline USD 5.00]

Please indicate in 3-5 sentences whether and why the conditions for the shortcut IRB apply to your experiment.

Our design involves:

No deception

Voluntary participation

No physical risks

No measurements of physical state (e.g., saliva samples, blood samples)

Clear information for subjects regarding duration, repetition, interactions and what is common information to subjects

Anonymity regarding who is interacting with whom during the experiment

Paying at least the opportunity cost of time, and avoid negative payments

Confidentiality of payments

Anonymous experimental data is kept separate from non-anonymous payments data.

Please specify your incentive structure.

Our outcome measure is hypothetical and we do not make use of additional incentives. We have added a statement with the aim of increasing the perceived consequentiality of the responses. Specifically, we would e-mail the results to Ivan Rudik, who is currently serving as Chief Environmental Economist at the White

House Office of Science and Technology Policy. If this is possible, we'd be happy to share screenshots of our e-mail with the participants.

Is the treatment effect identifiable using OLS, a probit/logit, or an ordered-logit model?

Yes.

What is the estimated length of the experiment? (excl. policy survey)

7.0 minutes.

Is your proposal using incentives?

No.

What is the average payment per participant (including flat fee + potential bonus payment)?

USD 2.00 (flat fee)

Which experimental software do you plan to use?

oTree.

What is your expected effect size? (Cohen's d for continuous outcome variables and Cohen's h for shares/proportions)

0.2